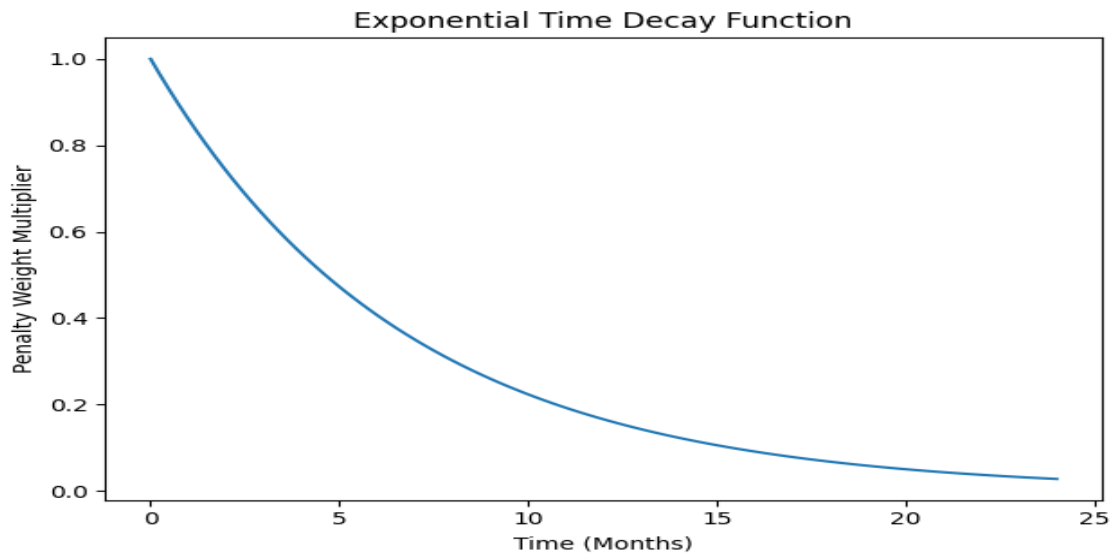


Agent Identity Registry (AIR)

Technical Supplement — Actuarial Risk Model (v2.0)

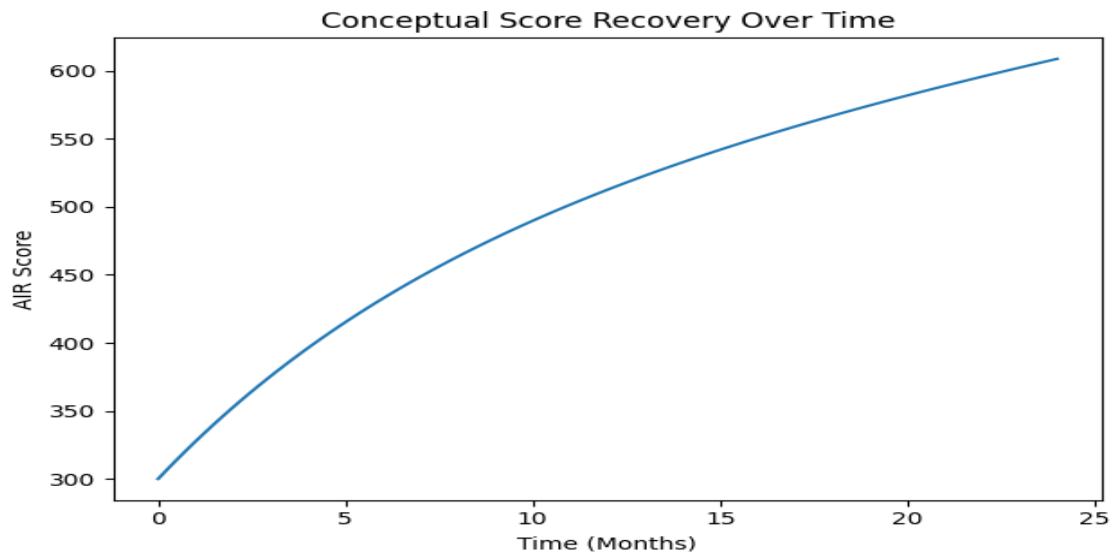
1. Time-Weighted Penalty Model

Negative events are adjusted using an exponential decay model to ensure historical accountability while enabling rehabilitation. The decay function follows: Adjusted Penalty = $P \times e^{(-\lambda t)}$, where λ is the decay constant and t is time elapsed.



2. Score Recovery Dynamics

AIR Scores evolve based on event-driven adjustments and sustained stability. Recovery is gradual and time-weighted, preventing abrupt manipulation while rewarding compliance.



3. Population Risk Distribution

AIR Score distribution across registered agents can be modeled using normal or skew-adjusted distributions for risk stratification, underwriting, and marketplace thresholding.

